

Ideal Cardiovascular Health is Inversely Associated with Incident Cancer: The Atherosclerosis Risk in Communities Study

Background The American Heart Association (AHA) has defined the concept of ideal cardiovascular health in promotion of their 2020 Strategic Impact Goals. We examined if adherence to ideal levels of the seven AHA cardiovascular health metrics was associated with incident cancers in the Atherosclerosis Risk In Communities (ARIC) study over 17-19 years of follow-up. **Methods and Results** After exclusions for missing data and prevalent cancer, 13,253 ARIC participants were included for analysis. Baseline measurements were used to classify participants according to seven AHA cardiovascular health metrics. Combined cancer incidence (excluding non-melanoma skin cancers) from 1987-2006 was captured using cancer registries and hospital surveillance; 2880 incident cancer cases occurred over follow-up. Cox regression was used to calculate hazard ratios for incident cancer. There was a significant ($p\text{-trend} < .0001$), graded, inverse association between the number of ideal cardiovascular health metrics at baseline and cancer incidence. Participants meeting goals for 6-7 ideal health metrics (2.7% of the population) had 51% lower risk of incident cancer than those meeting goals for 0 ideal health metrics. When smoking was removed from the sum of ideal health metrics, the association was attenuated with participants meeting goals for 5-6 health metrics having 25% lower cancer risk than those meeting goals for 0 ideal health metrics ($p\text{-trend} = .03$). **Conclusions** Adherence to the seven ideal health metrics defined in the AHA 2020 goals is associated with lower cancer incidence. The AHA should continue to pursue partnerships with cancer advocacy groups to achieve reductions in chronic disease prevalence.